npj | digital medicine **SUPPLEMENTARY INFORMATION**

Supplementary Information for:

Claimed quantum benefits in clinical machine learning do not survive symmetric evaluation

Sivaguru R. M.[1,*]

[1]Independent Researcher, Chennai, India. *e-mail: sivagurumurugan1@gmail.com

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1 Supplementary Note 1: The 17-Item Symmetric Evaluation Checklist & Protocol Algorithm

1.1 The 17-Item Protocol Checklist

The 17-item checklist operationalizes the seven protocol rules (P1–P7) into explicit, pre-specified binary criteria for auditing clinical AI studies claiming architectural advantage:

```
Section A: Calibration and Probability Evaluation (Rule P1)
[ ] Item 1:  Are predicted probabilities calibrated independently per arm?
             Rule: Each evaluated model must be calibrated separately using a dedicated
             validation/calibration partition prior to probability-based scoring.
[ ] Item 2:  Is calibration fitted strictly on validation/calibration data without test-set
    access?
             Rule: No test-set labels or features may be used during temperature scaling or
                 Platt scaling.
[ ] Item 3:  Are calibration metrics (ECE, Brier score) reported alongside discrimination (
    AUROC)?
             Rule: Studies claiming clinical utility must evaluate both probability fidelity
                  and ranking.

Section B: Optimization and Capacity Fairness (Rule P2)
[ ] Item 4:  Do classical and novel arms receive matched or equitable tuning budgets?
             Rule: The classical baseline must receive an equivalent number of
                 hyperparameter trials
             spanning standard model families.
[ ] Item 5:  Is hyperparameter selection performed via nested cross-validation or isolated
    validation splits?
             Rule: Test data must never influence architecture selection, learning rates, or
                  early stopping.
[ ] Item 6:  Are classical baselines competitive (including strong nonlinear models when
    tasks contain nonlinear structure)?
             Rule: Novel models must be compared against modern competitive nonlinear
                 baselines, not
             deliberately crippled or untuned linear baselines alone.
[ ] Item 7:  Are parameter counts, circuit depths, and convergence criteria explicitly
    reported?
             Rule: Model capacity, training time, and optimizer termination conditions must
                 be stated.

Section C: Partition Integrity and Leakage Prevention (Rule P3)
[ ] Item 8:  Are train, validation, and test splits strictly patient-disjoint?
             Rule: No patient records may cross partition boundaries.
[ ] Item 9:  Do repeated measurements (e.g., temporal EEG epochs, voice recordings, MRI
    slices) remain within their patient partition?
             Rule: Cross-validation must group by patient or subject identifier.
[ ] Item 10: Are all data-dependent transformations (PCA, scaling, imputations) fitted
    strictly inside training folds?
             Rule: Global dimensionality reduction on the full dataset prior to splitting is
                  prohibited.

Section D: Statistical Inference and Error Quantification (Rule P4)
[ ] Item 11: Are superiority claims supported by paired statistical hypothesis tests or
    difference intervals?
             Rule: Point estimate comparisons without uncertainty intervals or hypothesis
                 tests are invalid.
[ ] Item 12: Is repeated cross-validation analyzed using appropriate dependency corrections
    (e.g., Nadeau-Bengio)?
             Rule: Conventional t-tests that treat cross-validation folds as independent are
                  prohibited.
[ ] Item 13: Are multiple comparisons adjusted across the full family of evaluated model
    contrasts?
             Rule: Holm-Bonferroni or Benjamini-Hochberg multiplicity corrections must be
                 applied.
[ ] Item 14: Are equivalence claims supported by formal equivalence tests (TOST) rather than
     failure to reject the null?
             Rule: A non-significant p-value cannot be reported as proof of equivalence
                 without a pre-specified margin.

Section E: Transparency and Negative Result Reporting (Rules P5, P6, P7)
[ ] Item 15: Are all evaluated comparator arms and negative results fully reported without
    selective omission?
             Rule: Benchmarks must present the complete comparator suite, including
```

```
                underperforming configurations.
[ ] Item 16: Are all reported quantitative figures tied to immutable computational artifacts
     in a repository?
            Rule: Every table and figure must trace directly to saved prediction arrays or
                logs.
[ ] Item 17: Are stated physical or algorithmic mechanisms verified through ablation or code
    -level testing?
            Rule: Claims of quantum advantage mechanisms (e.g., entanglement) must be
                supported by
            ablation experiments demonstrating that the mechanism causally drives
                performance.
```

1.2 Formal Protocol Procedure

```
Algorithm S1: Symmetric Evaluation Protocol for Clinical Machine Learning
Input: Dataset D = {(x_i, y_i, g_i)} with patient/group identifiers g_i;
       Novel model family M_nov with hyperparameter space Theta_nov;
       Classical baseline families M_1, ..., M_K with spaces Theta_1, ..., Theta_K;
       Prespecified equivalence margin Delta_equiv; significance level alpha = 0.05.
Output: Paired generalization differences d_bar, dependency-corrected 95% CIs, p-values,
    verdict.

1. Partitioning at the Unit of Independence:
   Partition D into J x K outer folds grouped strictly by patient ID g_i such that
   intersect(g_train, g_test) = empty for every fold.
2. Leak-Free In-Fold Preprocessing:
   For each outer fold (j, k):
     Fit imputation, standard scaling, and dimensionality reduction strictly on D_train^(j,
         k).
     Apply fitted transformers to D_train^(j, k) and D_test^(j, k).
3. Symmetric Nested Hyperparameter Search:
   Within D_train^(j, k), partition into 3 inner folds.
   For each model family M_k:
     Evaluate all configs in Theta_k on inner validation folds.
     Select optimal configuration theta_k* = argmin NLL_inner(theta).
4. Independent Post-Hoc Calibration:
   Fit temperature scaling parameter T_k independently for each model on the inner
       validation split:
     T_k* = argmin NLL(y_val, p_val / T_k).
5. Held-Out Generalization Evaluation:
   Score frozen, calibrated models on untouched D_test^(j, k).
   Compute paired per-fold difference: d_(j, k) = Metric_quantum^(j, k) - Metric_classical^(
       j, k).
6. Dependency-Corrected Statistical Inference:
   Compute sample variance of differences S^2 = Var(d).
   Calculate Nadeau-Bengio corrected variance:
     sigma_NB^2 = (1 / (J * K) + n_test / n_train) * S^2.
   Derive test statistic t_NB = mean(d) / sigma_NB (df = J * K - 1).
   Compute corrected 95% confidence interval: mean(d) +/- t_(alpha/2, df) * sigma_NB.
7. Prespecified Decision Classification:
   - Superior: Lower bound of 95% CI > 0 (p < 0.05 favoring novel model).
   - Not Reproduced / Reversed: Upper bound of 95% CI < 0 (p < 0.05 favoring classical
       baseline).
   - Inconclusive: 95% CI crosses 0 with width exceeding practical decision thresholds.
   - Equivalent: 90% / 95% CI falls entirely inside [-Delta_equiv, +Delta_equiv] (p_TOST <
       0.05).
```

2 Supplementary Note 2: Multi-Class Chest Radiograph Triage Mapping & Bayes Optimality

2.1 Diagnostic Category Formulation

The multi-class diagnostic target $y \in \{0,1,2,3,4,5\}$ represents a clinical triage category derived deterministically from the predicted probabilities of eight radiological findings extracted by DenseNet-121 from frontal chest radiographs:

1. Pneumothorax (p_1)
2. Pleural Effusion (p_2)
3. Cardiomegaly (p_3)
4. Consolidation (p_4)
5. Atelectasis (p_5)
6. Pulmonary Edema (p_6)
7. Lung Lesion / Opacity (p_7)
8. Constant zero baseline reference channel ($p_8 \equiv 0.0$)

The clinical triage category is assigned via the priority decision rule:

$$y = \begin{cases} 1 \text{ (Pneumothorax)}, & \text{if } p_1 \geq 0.40 \\ 2 \text{ (Malignancy)}, & \text{else if } p_7 \geq 0.45 \\ 3 \text{ (Heart Failure)}, & \text{else if } p_3 \geq 0.50 \text{ or } p_6 \geq 0.45 \\ 4 \text{ (Pneumonia / Consolidation)}, & \text{else if } p_4 \geq 0.40 \\ 5 \text{ (COPD / Atelectasis)}, & \text{else if } p_5 \geq 0.50 \text{ or } p_2 \geq 0.50 \\ 0 \text{ (Normal / Low Risk)}, & \text{otherwise} \end{cases} \tag{1}$$

2.2 Separability and Theoretical Maximum

Because the mapping from probabilities $p \in [0,1]^8$ to $y \in \{0,\ldots,5\}$ is a deterministic partition of the input domain:

$$H(Y \mid X) = 0 \implies \text{Bayes Optimal Error} = 0\% \tag{2}$$

Any test misclassification by a fusion head is attributable to model capacity constraints, training error, optimizer convergence limitations, or feature extraction noise from DenseNet-121, rather than irreducible aleatoric label uncertainty. Furthermore, because this target is an algorithmic triage rule, model performance evaluates fidelity in emulating this diagnostic policy rather than direct clinical diagnostic efficacy or patient survival endpoints.

3 Supplementary Note 3: Analytic Derivation of the Interaction-Order Spectrum (Analysis M1)

3.1 Fourier Expansion of Variational Circuits

Consider an n-qubit quantum circuit where each classical feature $x_i \in [0,1]$ is encoded via single-qubit rotation $R_Y(\pi x_i) = \exp(-i\frac{\pi x_i}{2}Y)$. After an alternating sequence of variational entangling layers $U(\theta)$, the expectation value of a single-qubit Pauli-Z observable on qubit j can be expressed as:

$$f_j(x;\theta) = \langle 0^{\otimes n} | U^\dagger(x,\theta) Z_j U(x,\theta) | 0^{\otimes n} \rangle \tag{3}$$

By the generalized Fourier theorem for parameter-shift quantum circuits (Schuld et al., 2021), this expectation value is an exact trigonometric polynomial in x:

$$f_j(x;\theta) = \sum_{k \in \{-1,0,1\}^n} c_k(\theta) e^{i\pi k \cdot x} \tag{4}$$

The interaction order of a Fourier component indexed by $k = (k_1,\ldots,k_n)$ is the Hamming weight $\|k\|_0 = \sum_{i=1}^n \mathbb{I}(k_i \neq 0)$, representing the degree of multi-feature interaction.

3.2 Concentration of Energy in Entangled Circuits

For unentangled product states ($U(\theta) = \bigotimes_{i=1}^{n} R(\theta_i)$):

$$c_k(\theta) = 0 \quad \forall \, \|k\|_0 \geq 2 \tag{5}$$

In the 8-qubit variational circuit with circular CNOT entangling gates evaluated in Arm C1 (which incorporates the 7 clinical finding channels plus 1 constant-zero reference channel), numerical decomposition over the Boolean Pauli basis reveals:

$$E_1 = \sum_{\|k\|_0=1} |c_k|^2 = 0.0000 \times 10^{-4} \quad (0.000\% \text{ of non-trivial energy}) \tag{6}$$

$$E_{\geq 2} = \sum_{\|k\|_0 \geq 2} |c_k|^2 = 1.0000 \quad (100.000\% \text{ of non-trivial energy}) \tag{7}$$

Decomposing by individual interaction orders: order 0: 0.0%, order 1: 0.0%, order 2: 2.16%, order 3: 8.50%, order 4: 42.45%, order 5: 20.98%, order 6: 18.30%, order 7: 4.48%, order 8: 3.13%. Logit variance decomposition across $N = 56,534$ test patients confirms that 99.988% of model output variance is driven by interaction orders ≥ 2.

3.3 Reconciliation of 8-Qubit Model and 7-Channel Ablation (Analysis C1b)

In the 10-seed ablation isolating the 7 clinical finding channels without the constant reference qubit (Analysis C1b), the circuit was evaluated directly on $n = 7$ qubits. The entangled CNOT ring circuits similarly concentrated 100.0% of non-trivial coefficient energy into interaction orders ≥ 2 across all 10 seeds (mean interaction order: 4.3; plotted in Figure 2a), whereas unentangled product circuits strictly confined interaction energy to order 1 (0.0% in orders ≥ 2). Because the clinical triage rule is an additive threshold policy over individual radiological signs, the multi-qubit cross-terms generated by circular CNOT entanglement constitute unhelpful representational capacity that inflates variance without reducing bias, explaining the observed $+0.0104$-nat entanglement penalty.

4 Supplementary Note 4: Closed-Form Fidelity Kernel Reduction & Geometric Difference (Analysis M3)

4.1 Product-State Kernel Reduction

For an n-qubit product state feature map $|\psi(x)\rangle = \bigotimes_{i=1}^{n} |\phi(x_i)\rangle$ with $|\phi(x_i)\rangle = \cos(\pi x_i/2)|0\rangle + \sin(\pi x_i/2)|1\rangle$, the quantum fidelity kernel is:

$$K_Q(x,x') = |\langle\psi(x)|\psi(x')\rangle|^2 = \prod_{i=1}^{n} |\langle\phi(x_i)|\phi(x_i')\rangle|^2 \tag{8}$$

Evaluating the inner product analytically:

$$\langle\phi(x_i)|\phi(x_i')\rangle = \cos\left(\frac{\pi x_i}{2}\right)\cos\left(\frac{\pi x_i'}{2}\right) + \sin\left(\frac{\pi x_i}{2}\right)\sin\left(\frac{\pi x_i'}{2}\right) = \cos\left(\frac{\pi(x_i - x_i')}{2}\right) \tag{9}$$

Squaring yields the closed-form classical cosine product kernel:

$$K_Q(x,x') = \prod_{i=1}^{n} \cos^2\left(\frac{\pi(x_i - x_i')}{2}\right) \tag{10}$$

This establishes that quantum fidelity kernels without entanglement are mathematically identical to classical product kernels, eliminating any theoretical rationale for quantum advantage in product circuits.

4.2 Geometric Difference on Clinical Embeddings

Following Huang et al. (2021), the regularized geometric difference between a classical kernel K_1 and a quantum kernel K_2 is:

$$g_\lambda(K_1 \parallel K_2) = \sqrt{\|\sqrt{K_2}(K_1 + \lambda I)^{-1}\sqrt{K_2}\|_\infty} \tag{11}$$

When $g_\lambda \ll \sqrt{N}$, classical kernel methods achieve comparable generalization performance to the quantum kernel. Evaluated on 2,000 samples from the CheXpert development cohort with regularization $\lambda = 10^{-3}$ and median bandwidth $\gamma = 3.734$:

$$g_\lambda(K_{\text{RBF}} \parallel K_Q) = 1.2712 \ll \sqrt{N} \approx 44.72 \tag{12}$$

This confirms that the classical RBF kernel readily learns any function in the reproducing kernel Hilbert space (RKHS) of the quantum kernel on this clinical feature distribution.

5 Supplementary Note 5: Full 17-Item Audit Scoring Matrix for Eligible Literature Studies

Supplementary Table 1 | Complete 17-item checklist scores across six assessable published digital health QML studies.

Checklist Item	Parthasarathi (CKD)	Daga (Parkinson/Pima)	Aksoy (EEG)	Akpinar (Glioma)	Ghisoni (Heart)	Jahin (Heart)
1. Independent Arm Calibration	NO	NO	NO	NO	NO	NO
2. Validation-Only Calibration	NO	NO	NO	NO	NO	NO
3. Calibration Metrics Reported	NO	NO	NO	NO	NO	NO
4. Matched Classical Tuning Budget	NO	NO	NO	NO	NO	NO
5. Nested CV / Isolated Selection	NO	NO	NO	NO	NO	NO
6. Competitive Nonlinear Baselines	NO	YES	NO	NO	NO	NO
7. Explicit Parameters / Convergence	YES	YES	YES	YES	YES	YES
8. Patient-Disjoint Partitioning	YES	NO	NO	YES	YES	YES
9. Grouped Repeated Measures	N/A	NO	NO	N/A	N/A	N/A
10. In-Fold Feature Preprocessing	NO	NO	NO	NO	YES	NO
11. Paired Inferential Testing	NO	NO	NO	NO	NO	NO
12. Dependency Correction (Nadeau-Bengio)	NO	NO	NO	NO	NO	NO
13. Multiplicity Adjustment	NO	NO	NO	NO	NO	NO
14. Formal Equivalence Testing (TOST)	NO	NO	NO	NO	NO	NO
15. Complete Outcome Reporting	YES	YES	YES	YES	YES	YES
16. Immutable Derived Artifacts	NO	NO	NO	NO	NO	NO
17. Mechanistic Code Verification	NO	NO	NO	NO	NO	NO
Total Compliant Items (/17)	**3 / 16**	**3 / 17**	**2 / 17**	**3 / 16**	**4 / 16**	**3 / 16**

6 Supplementary Note 6: Code Audit Findings and Artifact Discrepancy Ledger

6.1 CheXpert Prototype Historical Audit

Auditing the internal clinical decision-support prototype repository revealed several historical evaluation asymmetries:

1. **One-Sided Temperature Scaling:** Temperature scaling was initially applied exclusively to the quantum circuit, yielding an artificial ECE gap (0.020 vs 0.276). Applying temperature scaling symmetrically reduced the ECE gap by $> 73\%$.
2. **Unconverged Classical Baselines:** The initial linear baseline was trained with a fixed learning rate of 0.001 for only 30 epochs without early stopping, resulting in underfitting. Standardization of optimizer schedules eliminated all statistically significant performance gaps on the development cohort.
3. **Silent Constant Reference Channel:** An 8th channel was passed into the circuit set permanently to 0.0, acting as an unrotated ground-state reference qubit.

7 Supplementary Note 7: Hyperparameter Search Spaces, Compute Budgets, and Convergence

7.1 Analysis C2 Search Grid Specification

In Analysis C2, each model was tuned across an identical budget of 36 configurations:

- **Variational Quantum Circuit (VQC, 36 configs):**
  - Depth $L \in \{1,2,3,4,6\}$ layers.
  - Encoding rotation scale $s \in \{0.5,1.0,2.0\}$.
  - Initialization standard deviation $\sigma \in \{0.01,0.1,1.0\}$.
  - Learning rate $\eta \in \{0.01,0.05,0.1\}$.

  Winning Selected Model: $L = 4$ layers, scale $s = 0.5$, initialization $\sigma = 0.01$, learning rate $\eta = 0.01$. Total parameters: **118** (64 angles +54 readout weights). Calibration NLL on $n = 1,600$: 1.1355; calibration temperature $T = 1.1354$. Test NLL on 56,534 held-out patients: **1.1144**.
- **Product-of-Experts (PoE, 36 configs):**
  - Learning rate $\eta \in \{0.003,0.01,0.05,0.1,0.3\}$.
  - Weight decay $\lambda \in \{0.0,10^{-5},10^{-4},10^{-3}\}$.
  - Initialization standard deviation $\sigma \in \{0.01,0.1,1.0\}$.

  Winning Selected Model: Learning rate $\eta = 0.05$, weight decay $\lambda = 10^{-4}$, initialization $\sigma = 0.01$. Total parameters: **54**. Calibration NLL on $n = 1,600$: 1.1369; calibration temperature $T = 1.0751$. Test NLL on 56,534 held-out patients: **1.1170**.
- **Attention-Gated Nonlinear Head (36 configs):**
  - Learning rate $\eta \in \{0.003,0.01,0.05,0.1,0.3\}$.
  - Weight decay $\lambda \in \{0.0,10^{-5},10^{-4},10^{-3}\}$.
  - Initialization standard deviation $\sigma \in \{0.01,0.1,1.0\}$.

  Winning Selected Model: Learning rate $\eta = 0.1$, weight decay $\lambda = 10^{-4}$, initialization $\sigma = 0.01$. Total parameters: **126**. Calibration NLL on $n = 1,600$: 1.1357; calibration temperature $T = 1.0743$. Test NLL on 56,534 held-out patients: **1.1155**.

7.2 Computational Runtime and Wall-Clock Asymmetry

We logged wall-clock training and validation runtimes across all 36 configurations per arm on identical host hardware:

- **Product-of-Experts (PoE):** 161.4 seconds total (mean 4.5 s / config; range 1.2–14.3 s).
- **Attention-Gated Head:** 228.1 seconds total (mean 6.3 s / config; range 2.1–14.7 s).
- **Quantum Variational Circuit (VQC):** 14,969.4 seconds total (4.16 hours; mean 415.8 s / config; range 114.6–1642.6 s).

Because quantum statevector simulation involves multi-qubit tensor contractions rather than simple linear matrix products, the quantum circuit consumed approximately **92.8 times more wall-clock compute** than classical PoE and 65.6 times more than the Gated Head. We emphasize that while the number of evaluated configurations ($k = 36$) was strictly matched, computational training budgets differed by nearly two orders of magnitude.

7.3 C1 versus C2 PoE Comparison and Selection Variance

In Analysis C1, the reference Product-of-Experts baseline was evaluated under default matched settings ($\eta = 0.01, \lambda = 0, \sigma = 0.1$ chosen from a 3-point learning-rate grid), achieving test NLL **1.1157** on the 56,534 held-out patients ($T = 1.1017$). In Analysis C2, PoE was granted an expanded 36-configuration search space. The configuration selected by minimum NLL on the 1,600 development calibration cases was configuration 17 ($\eta = 0.05, \lambda = 10^{-4}, \sigma = 0.01$), which achieved marginally lower calibration NLL (1.1369 vs 1.1371), but yielded test NLL **1.1170** on the 56,534 held-out patients. The $+0.0013$-nat difference between C2 PoE (1.1170) and C1 PoE (1.1157) is an out-of-sample test-set evaluation difference on the 56,534 held-out patients between two distinct model configurations: selecting configuration 17 on the finite development calibration split ($n = 1,600$) produced a marginal calibration gain that did not generalize to lower loss on the separate 56,534-patient confirmatory cohort.

The primary C2 contrast evaluates models under the exact same 36-configuration search budget ($\Delta\text{NLL} = 1.1144 - 1.1170 = -\mathbf{0.00260}$ nats; $t = -5.10, p = 3.07 \times 10^{-7}$, $p_{\text{TOST}} < 10^{-6}$). When compared instead against the fixed C1 default PoE baseline (1.1157), the difference shrinks to $\Delta\text{NLL} = -0.0013$ nats. Both contrasts fall strictly within the prespecified operational equivalence margin (± 0.010 nats).

7.4 Development Calibration Reuse vs Confirmatory Independence

During development, the 1,600 calibration patients were used both for early-stopping checkpoint selection (monitoring validation loss with patience of 20 epochs) and for fitting the scalar Platt temperature scaling parameter T^* via L-BFGS. No model was refit on the pooled development set prior to calibration. While this dual use introduces a minor optimistic selection bias on development validation loss, the confirmatory evaluation cohort of 56,534 patients was completely sealed, preserved zero patient overlap ($|S_{\text{dev}} \cap S_{\text{test}}| = 0$), and was evaluated only once after all configurations, weights, and calibration temperatures were entirely frozen.

8 Supplementary Note 8: Statistical Formulations and Inferential Distinctions

8.1 Inconclusive Comparisons vs. Formal Equivalence Testing: Biostatistical Distinction

A fundamental principle of clinical biostatistics is that "absence of evidence is not evidence of absence" (Altman & Bland, 1995). A common misinterpretation in machine learning benchmarks is to treat a non-significant superiority test ($p > 0.05$) as proof that two methods are equivalent. This assertion is mathematically invalid unless evaluated using formal equivalence testing (such as Two One-Sided Tests) backed by adequate statistical power to rule out practically meaningful differences.

This distinction is prominently illustrated in our literature benchmark evaluations. For instance, in the low-data heart disease benchmark (Ghisoni et al., 2026; PMID 41513819), the original study evaluated a 1-layer quantum neural network against classical baselines on a 10% training fraction ($n_{\text{train}} = 74, n_{\text{test}} = 184$ per outer fold across 25 outer folds of 5×5 repeated CV). Under validation-based early stopping, the observed point difference was $\Delta = -1.10\%$. When analyzed using a conventional paired Student's t-test (which erroneously treats CV folds as independent), the 95% CI appears deceptively narrow: $[-3.20\%, +1.00\%]$ ($p = 0.310$). However, when evaluated under the rigorous Nadeau–Bengio correction accounting for outer-fold data overlap with the actual low-data ratio $n_{\text{test}}/n_{\text{train}} = 184/74 = 2.486$, the corrected standard error widens significantly:

$$\text{SE}_{\text{NB}} = \sqrt{\left(\frac{1}{25} + 2.486\right) S^2} = \sqrt{2.526} \cdot S \approx 1.589 \cdot S \tag{13}$$

yielding a corrected 95% confidence interval of -18.45 **to** $+16.25$ **percentage points** ($t = -0.13, df = 24, p = 0.897$).

An uncertainty interval of nearly 35 percentage points spans substantial clinical superiority in either direction. Consequently, this outcome supports an **inconclusive result**, not proof that the methods are equivalent. The benchmark lacked the sample size to detect whether the models differ by 5, 10, or 15 percentage points. Similar wide uncertainty intervals characterize other external benchmarks evaluated under dependency correction (e.g., Parkinson's Voice 95% CI: $[-24.97\%, +25.64\%]$).

In stark contrast, our confirmatory evaluation on CheXpert was powered on **56,534 independent patients**. The resulting 95% confidence interval was extremely narrow ($[-0.00359, -0.00160]$ nats), lying entirely inside our pre-specified operational equivalence band (± 0.010 nats) and rejecting non-equivalence via Two One-Sided Tests ($p_{\text{TOST}} < 10^{-6}$). The paper carefully distinguishes between these two regimes: external tasks were statistically inconclusive, whereas the confirmatory trial formally confirmed practical equivalence.

8.2 Operational Equivalence Margin Rationale and Justification

In Analysis C2, the equivalence margin was pre-specified as $\Delta_{\text{equiv}} = \pm 0.010$ nats. Mathematically, an average difference in negative log-likelihood of 0.010 nats bounds the population expected cross-entropy difference:

$$\Delta\text{NLL} \leq 0.010 \iff \mathbb{E}_{(x,y)}\left[\log \frac{p_{\text{classical}}(y \mid x)}{p_{\text{quantum}}(y \mid x)}\right] \leq 0.010 \text{ nats} \tag{14}$$

which corresponds to an average predictive log-loss divergence of 0.010 nats per observation across the cohort. While $\exp(0.010) \approx 1.01005$ reflects the ratio of geometric-mean predicted likelihoods under an average 0.010-nat divergence, an average negative-log-likelihood difference across a cohort does **not**, by itself, mathematically bound individual-patient risk ratios or worst-case clinical risk for specific subgroups.

Operational Status and Absence of Clinical Validation: This margin was chosen strictly as an **operational statistical benchmark** prior to test unblinding to define a narrow band of predictive indistinguishability on negative log-likelihood. It has **not been clinically validated** through prospective patient trial outcomes, physician decision audits, or clinical risk-benefit trade-off consensus. In clinical regulatory frameworks (such as FDA or EMA non-inferiority trials), clinical non-inferiority margins require formal stakeholder consensus linking model error directly to tangible patient outcomes (e.g., excess mortality, delayed therapy, or diagnostic misclassification). Unless supported by such empirical clinical validation, the ± 0.010-nat threshold must remain explicitly designated as an operational statistical margin.

To demonstrate that the conclusion of practical equivalence does not depend arbitrarily on this operational threshold, we conducted a sensitivity analysis across four alternative operational margins:

Supplementary Table 2 | Equivalence testing sensitivity across alternative operational margins (Analysis C2).

Operational Margin Δ_{equiv}	**Contrast vs PoE** ($\Delta = -0.0026$)	p_{TOST} **(PoE)**	**Contrast vs Gated** ($\Delta = -0.0012$)	p_{TOST} **(Gated)**
± 0.020 nats	Equivalent	$p < 10^{-15}$	Equivalent	$p < 10^{-15}$
± 0.010 nats (Pre-specified)	Equivalent	$p = 1.84 \times 10^{-7}$	Equivalent	$p = 6.42 \times 10^{-11}$
± 0.005 nats	Equivalent	$p = 3.88 \times 10^{-4}$	Equivalent	$p = 1.05 \times 10^{-5}$
± 0.002 nats	Non-Equivalent	$p = 0.884$	Equivalent	$p = 0.013$

Note: Margins are operational average-NLL statistical benchmarks ($\mathbb{E}[\Delta\ell] \leq \Delta_{equiv}$); an average NLL difference does not bound individual-patient risk ratios.

8.3 Scope and Generalizability Boundaries

The empirical evidence provided by our five reimplemented publications and six clinical tasks supports a carefully bounded conclusion: published claims of quantum advantage in these specific digital health settings collapse when subjected to symmetric optimization, independent calibration, and leak-free validation. However, this finding **cannot be generalized to a universal claim** that quantum computing can never offer an advantage in machine learning or clinical healthcare.

The empirical boundary of our work is explicitly defined by:

1. **Model Families:** Shallow hardware-efficient variational circuits (VQC, QNN, 4–12 qubits) and fidelity quantum kernels.
2. **Data Encodings:** Single-qubit Pauli-Y angle encodings $R_Y(\pi x_i)$.
3. **Data Representations:** Tabular clinical features and classical deep CNN feature embeddings.
4. **Computational Paradigm:** Exact classical statevector simulation.

Alternative quantum paradigms—including fault-tolerant algorithms with asymptotic polynomial or exponential speedups (e.g., HHL algorithm, quantum phase estimation), native quantum states from quantum sensors, multi-qubit interactive Hamiltonian feature maps that cannot be classically simulated, and biological domains where physical classical simulation is intractable (e.g., molecular dynamics, drug-target binding, and quantum chemistry)—fall entirely outside our experimental boundary and may hold genuine quantum potential.

9 Supplementary Note 9: Confirmatory Master Results Ledger & Literature Benchmarks

9.1 Confirmatory Trial Results

Supplementary Table 3 | Complete results across confirmatory analyses on $N = 56,534$ patients.

Analysis Arm / Target	Point Estimate	95% Confidence Interval	Test Statistic / p-value	Pre-Registered
C1: Matched Default (ΔNLL vs PoE)	$+0.0205$ nats	$[+0.01858, +0.02240]$	$t = 21.034$, $p = 7.65 \times 10^{-98}$	Quantum Materia
C1: Uncertainty Sensitivity (y_{u1})	$+0.0074$ nats	$[+0.00617, +0.00868]$	$t = 9.871$, $p = 6.12 \times 10^{-23}$	Quantum W
C1b: Entanglement Penalty (ΔNLL)	$+0.0104$ nats	$[+0.00934, +0.01138]$	$p_{\text{boot}} = 0.00128$	Penalty Repl
C2: Tuned Quantum vs PoE (ΔNLL)	-0.0026 nats	$[-0.00359, -0.00160]$	$p = 3.07 \times 10^{-7}$, $p_{\text{TOST}} < 10^{-6}$	Practically Equ
C2: Tuned Quantum vs Gated (ΔNLL)	-0.0012 nats	$[-0.00192, -0.00041]$	$p = 0.00247$, $p_{\text{TOST}} < 10^{-6}$	Practically Equ
M2: Order-1 Classical Fourier (ΔNLL)	-0.0238 nats	$[-0.0257, -0.0219]$	$p < 10^{-4}$	Classical Fouri
M2: Order-2 Classical Fourier (ΔNLL)	-0.0267 nats	$[-0.0286, -0.0248]$	$p < 10^{-4}$	Classical Fouri

9.2 Literature Benchmark Detailed Master Ledger

Supplementary Table 4 | Complete statistical ledger of literature re-implementations across 25 outer folds.

Benchmark Task	Novel Arm	Classical Baseline	Symmetric Margin Δ	Corrected 95% CI	p-value (NB)	Status
Kidney Disease (CKD)	VQC (8q)	Tuned Ensemble	-9.65%	$[-14.92\%, -4.38\%]$	$p = 0.000915$	Not Reproduced*
Parkinson Voice	QKL Stacking	Tuned Linear	$+0.33\%$	$[-24.97\%, +25.64\%]$	$p = 0.979$	Inconclusive
Pima Diabetes	QKL Stacking	Tuned RBF	-5.30%	$[-11.00\%, +0.39\%]$	$p = 0.0666$	Inconclusive
Schizophrenia EEG	QSVM	SVM-RBF	-7.60%	$[-23.01\%, +7.81\%]$	$p = 0.3187$	Partition Artifact
Glioma Grading (TCGA)	VQC (5q)	Tuned GBDT	$+0.19\%$	$[-1.22\%, +1.60\%]$	$p = 0.782$	Inconclusive
Glioma BraTS 2020	QKL (10q)	SVM-RBF	-0.0217 (AUROC)	$[-0.0543, +0.0109]$	$p = 0.191$	Inconclusive
Heart Disease (Low-Data)	QNN (11q)	Tuned MLP	-1.10%	$[-18.45\%, +16.25\%]$	$p = 0.897$	Inconclusive

*The published 99.2% accuracy claim utilized an under-specified autoencoder and Kalman filter that could not be replicated from the published text; on the published 13 LASSO features, our reimplementation reached only 83.5%, while classical models achieved 96–98%. Under symmetric nested tuning, classical models maintained significant superiority ($\Delta = -9.65\%$, $p = 0.0009$).